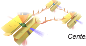

The Road to the Functional Quantum Network

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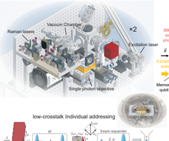
Introduction

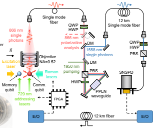
Quantum networks and repeaters represent a leading pathway toward scaling quantum information systems, underpinning applications such as distributed quantum computing, long-distance secure communication and blind quantum computing. The implementation of the functional quantum network is bottlenecked by the inherent loss and decoherence over long distances. In our works, we overcome these challenges by implementing a hybrid temporal multiplexing scheme within a dual-type qubit encoding framework over a metropolitan distance, simultaneously emphasizing high fidelity. This collective body of works realizes all critical pillars required for the robust, metropolitan-scale quantum network for practical applications.

Experimental Apparatus

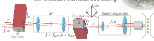
Platform overview of single node


Raman lasers
Vaccum Chamber
×2
Excitation laser
Single photon objective

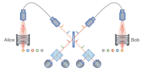
Overall scheme of single node


Single mode fiber
12 km
Single mode fiber
866 nm single photons
QWP
HWP
866 nm polarization analysis
DM
1558 nm single photons
QWP
HWP
PBS
$\vec{B}$
Excitation laser
Objective NA=0.52
DM
1950 nm pumping
PBS
SNSPD
Raman lasers
Memory qubit
Comm. qubit
HWP
PPLN waveguide
729 nm addressing lasers
FPGA
E/O
12 km fiber
E/O

low-crosstalk Individual addressing


Z
729 nm
f, ϕ
f_{RF}, ϕ_{RF}
X
+1
4f
$f + f_{RF}, \phi + \phi_{RF}$
f_{RF}, ϕ_{RF}
Y
-1
Beam expander
f, ϕ

Bell State Measurement apparatus

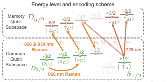


Elements for our functional quantum network node

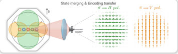
- Linear Paul trap with ^{40}Ca^{+}
- Dissipative lasers for state preparation and measurement
- Coherent control for quantum information process
- Individual addressing for flexible qubits controlling
- Robust excitation scheme for heralding entanglement
- High NA objective guiding emitted photon into optical link
- Sagnac quantum frequency conversion
- High-speed distributed electrionic control

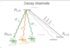
Scheme Outline

Energy level and encoding scheme


Memory Qubit Subspace
$D_{5/2}$
-5/2
$|\downarrow_C\rangle$
-3/2
$|\uparrow_C\rangle$
-1/2
+1/2
+3/2
$|\downarrow_M\rangle$
+5/2
$|\uparrow_M\rangle$
850 & 854 nm Raman
T1
T2
T3
T3
T2
T1
729 nm
Commun. Qubit Subspace
-3/2
$|2_C\rangle$
-1/2
$|0_C\rangle$
+1/2
$|1_C\rangle$
+3/2
$D_{3/2}$
866 nm Raman
-1/2
$|\downarrow_S\rangle$
+1/2
$|\uparrow_S\rangle$
$S_{1/2}$

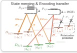
Adopting the dual-type qubit architecture orthogonally segregates the state space into dedicated **Memory** and **Communication** subspaces. The 729nm and Raman lasers serve as the essential coherent link, enabling internal state transfer and entangling gates.


State merging & Encoding transfer
B
780HP
$\sigma \rightarrow H$ pol.
$\pi \rightarrow V$ pol.


Decay channels
$P_{1/2}$
σ_+
π
σ_-
$D_{3/2}$
$S_{1/2}$
$|2_C\rangle$
$|0_C\rangle$
$|1_C\rangle$
$\sqrt{1/2}$
$-\sqrt{1/3}$
$\sqrt{1/6}$

By encoding the quantum information into the photon's **polarization degree-of-freedom**, we achieve the heralded generation of entanglement between the stationary qubit (ion) and the flying qubit (photon):

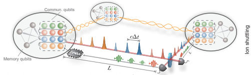
$$|\phi^+_{IP}\rangle = \frac{1}{\sqrt{3}}|0_C\rangle|\pi\rangle + \frac{1}{\sqrt{6}}|1_C\rangle|\sigma_-\rangle + \frac{1}{\sqrt{2}}|2_C\rangle|\sigma_+\rangle$$

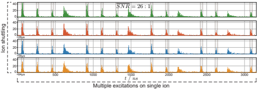

State merging & Encoding transfer
$P_{1/2}$
Δ
$\Delta = 18GHz$
854nm
850nm
866nm
σ_+
σ_-
$D_{5/2}$
$|\downarrow_C\rangle$
$|\uparrow_C\rangle$
$|\downarrow_M\rangle$
$|\uparrow_M\rangle$
Polarization protected
$D_{3/2}$
$|2_C\rangle$
$|0_C\rangle$
$|1_C\rangle$

Following the efficient collection of the single photon into the optical fiber, a crucial entanglement mapping operation (**state merging** and **encoding transfer**) is executed internally. This process projects the final state into a **Bell state**:

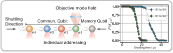
$$|\phi^+_{IP}\rangle = \frac{1}{\sqrt{2}}|\downarrow_C\rangle|V\rangle + \frac{1}{\sqrt{2}}|\uparrow_C\rangle|H\rangle$$

Metropolitan-Scale Hybrid-Multiplexed Ion-Photon Entanglement

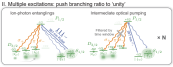

Commun. qubits
Memory qubits
$c\Delta t$
L
L



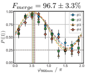
I. Ion shuttling: move commun. qubit to photonic interface

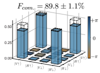

Objective mode field
Shuttling Direction
Commun. Qubit
Memory Qubit
#4
#3
#2
#1
Individual addressing
$P(S_{1/2})$
S1 to S4
S1 to S2
Shuttling time / μs

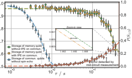
II. Multiple excitations: push branching ratio to 'unity'


Ion-photon entanglings
$P_{1/2}$
397 nm Pulse
$D_{3/2}$
$S_{1/2}$
Intermediate optical pumping
Filtered by time window
397 nm
× N

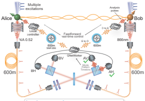
- The histogram of totally **44 modes** is shown, with multiplexing enhancement factor **16-fold** compared with the single-mode case.
- The stored quantum information in the **memory qubit is protected** from the dissipative operations on the commun. qubit via **dual-type encoding**, and spontaneous emission error in the memory qubit can be detected via a **mid-circuit** meansurement.
- **Link efficiency 116% over 12km** optical fiber, with the inherent survival rate of D state fully taken into account.
- All the **mode is unified** by resorting dynamic adjustment with 96.7% merging fidelity and 89.8% bell state fidelity.







Record-High-Fidelity Multiplexed Ion-Ion Entanglement Over 1.2km


Multiple excitations
Alice
Analysis pulses
Bob
Local controller
Feedforward real-time control
O to E
E to O
NA 0.52
600m SMF28E
600m
866nm
QNetWorker
BV
AV
BH
AH
Mode Pattern Timebins
600m
600m
QWP
HWP
Objective
50:50 NPBS
780HP
PBS
Fiber coupler

The joint ion-photon states of Alice and Bob is decomposed into a combination of different Bell states:

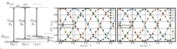
$$|\Psi\rangle = |\phi^+_{IP}\rangle_A \otimes |\phi^+_{IP}\rangle_B = 1/2\left(|\phi^+_{II}\rangle \otimes |\phi^+_{PP}\rangle + |\phi^-_{II}\rangle \otimes |\phi^-_{PP}\rangle + |\psi^+_{II}\rangle \otimes |\psi^+_{PP}\rangle + |\psi^-_{II}\rangle \otimes |\psi^-_{PP}\rangle\right)$$

Event (AH, AV) heralds product entangled ion-photon state collapses into the desired ion-ion entangled state:

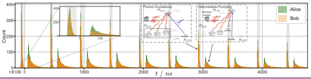
$$|\psi^+_{II}\rangle = \alpha\,|0_C\rangle_A \otimes \left(\sqrt{1/4}|1_C\rangle_B + \sqrt{3/4}e^{i\phi^B_M}|2_C\rangle_B\right) + \beta e^{i\phi_E}\left(\sqrt{1/4}|1_C\rangle_A + \sqrt{3/4}e^{i\phi^A_M}|2_C\rangle_A\right) \otimes |0_C\rangle_B$$

Intermodular phase in the ion-ion entangled state can be directly determined via the ion-photon entanglement according the theoretical analysis:

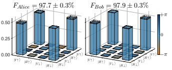
$$\phi_E(t) - \left[\phi^A_e(z_H,t) - \phi^B_e(z_H,t)\right] = -\Delta\omega_{e0}\Delta\tau + \Delta k_0(z_H - z_V) \approx 2\pi \times 10^{-5}$$



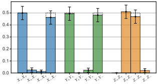
- Arrival time difference ~**6ps**
- **10 temporal** photonic modes, **4.59-fold** speedup



- Here we use a **strict heralding protocol** that both the flying qubits (from local nodes to the center station) and the heralding signals (from the center station back to each node) are transmitted through the long fibers to simulate the round-trip communication time in the **real-world scenario**.

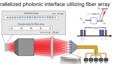


- Bell state fidelity **95.9 ± 1.5% over 1.2km** optical channel
- Fidelity beyond **99% is reachable** in **polarization-encoded scheme**



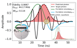
Futural Work

- Parallelized photonic interface utilizing fiber array

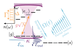

Full-atom array
4.7 μm
Five-site array for fibre array
20 μm
λ = 1,089 nm
3.8 μs

N pair entanglement with N/η time consumed during one shot.

- Unified photonic interface of quantum network node

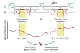


C. Thomas et al. *arXiv* (2025).

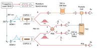

ω
Γ
Δ
$|e_\omega\rangle$
$n(\omega)$
$g\sqrt{N}$
$|g\rangle$
κ
$\mathcal{E}_{in}$
$\mathcal{E}_{out}$
$\Omega(t)$
$|s\rangle$

P. Cussenot et al. *arXiv* (2025).

Arbitrary control of the temporal waveform of photons in spontaneous emission and cavity QED.



A. Ruskuc. et al. *Nature* (2025).



T. M. Zhao. et al. *PRL* (2014).

Arbitary control of the phase feedforward compensation in heterodyning system.

References

[1] P. C. Lai, Y. Wang, J. X. Shi, et al. Realization of a crosstalk-free two-ion node for long-distance quantum networking. *Phys. Rev. Lett.* 134, 070801 (2025).

[2] Z. B. Cui, et al. Metropolitan-scale ion-photon entanglement via a quantum network node with hybrid multiplexing enhancements. *Nat. Commun* (2026).

[3] Z. B. Cui, Z. Q. Wang, et al. Multiplexed ion-ion entanglement over 1.2 kilometer fibers. *Phys. Rev. Lett. In Review* (2026).

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